$$L = -[y \log(a) + (1-y)\log(1-a)]$$

where,

$$a = \sigma(z) = \frac{1}{1-e^{-z}}$$

$$\frac{\partial L}{\partial a} = -\left[y \frac{\partial}{\partial a}\log(a) + (1-y)\frac{\partial}{\partial a}\log(1-a)\right]$$

$$= -\left[y \cdot \frac{1}{a} + (1-y)\cdot\frac{-1}{1-a}\right]$$

$$= -\frac{y}{a} + \frac{1-y}{1-a}$$

$$= \frac{-y + ay + a - ay}{a(1-a)}$$

$$= \frac{a-y}{a(1-a)}$$

$$\therefore \frac{\partial L}{\partial a} = \frac{a-y}{a(1-a)}$$

$$\frac{\partial a}{\partial z} = a(1-a)$$

Using chain rule,

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z}$$

$$= \left(\frac{a-y}{a(1-a)}\right) \cdot a(1-a)$$

$$= a-y$$

$$\therefore \frac{\partial L}{\partial z} = a-y$$